Jacob Sanders

Full Stack Software Developer

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SUMMARY

Full-stack developer with 13+ years building products where software meets physical hardware. Nine of those years were spent in real-time embedded systems and firmware, covering Bluetooth Low Energy, USB and serial communication, and the host-to-device protocols between them, from board bring-up through production release. The last several have gone to the application layer on top: Flutter/Dart cross-platform apps, React and Next.js web apps, and the REST and GraphQL services, relational schemas, and cloud infrastructure behind them. Has worked alongside mechanical, electrical, manufacturing, and QA engineers for most of his career, including as primary liaison to Apple's MFi program office. Comfortable debugging across the software, firmware, and hardware boundary, and used to owning a product from prototype through production.

SYSTEMS & HARDWARE INTEGRATION

- **Bluetooth / BLE:** Bluetooth Low Energy firmware for the Nordic nRF52832 SoC, including the host-to-device communication protocol; architected embedded firmware across a portfolio of Bluetooth audio products
- **USB & serial communication:** C# desktop application performing USB firmware updates on consumer audio hardware; USB Mass Storage optimization cutting firmware upgrade time from ∼3 min to ∼60 sec
- **Embedded systems:** 9 years as Senior Firmware Engineer and Firmware Architect, taking consumer audio and industrial scanning hardware from board bring-up through production release
- **Firmware communication & device protocols:** host-to-device protocol design; an embedded JavaScript engine that let customers reconfigure device behavior in the field without shipping a firmware release
- **Real-time systems:** real-time firmware for barcode scanning and consumer audio products; Linux kernel drivers for image sensor and GPIO/PWM timing

EXPERIENCE

Lien Into Revenue Services, LLC — Remote
Co-Founder / Lead Engineer — August 2025 – Present

- Build and maintain a production full-stack SaaS platform end-to-end, a Next.js/React/TypeScript frontend on a serverless AWS backend (Lambda, AppSync GraphQL, DynamoDB, Cognito), covering the full case lifecycle from intake through settlement collection
- Design the data models, REST and GraphQL API surfaces, authentication and user accounts, and role-based multi-tenant access rules; deploy as TypeScript infrastructure-as-code with per-branch environments
- Hold the quality line with Jest and React Testing Library unit tests, Playwright end-to-end suites, code review, and CI/CD, then carry each feature through release and post-release support

ARK Electronics — Remote
Full-Stack Engineer (1099 Contract) — February 2025 – April 2025

- Embedded with the client to scope requirements from scratch and ship a cloud-based manufacturing traceability platform end-to-end in weeks, covering product setup, serial number generation, barcode artifact creation, and test-result tracking for production and QA workflows
- Built the React/TypeScript frontend and a serverless AWS backend exposing dual API surfaces, GraphQL for the web UI and REST for external clients, with API key management, throttling, and event-driven job status updates
- Translated shop-floor process from manufacturing and QA staff into working software, adjusting scope as the client's requirements sharpened during the build

The Honeydew App, Inc. — Salt Lake City, UT
Founder — April 2023 – February 2025

- Built a cross-platform consumer mobile application in **Flutter and Dart** on an MVVM architecture, covering the full user flow from onboarding through job booking and dispatch
- Designed the serverless TypeScript backend on Supabase/PostgreSQL: the relational schema, authentication and user accounts, row-level access rules, and real-time job dispatch between customers and service providers
- Owned all system architecture and the majority of engineering while directing 4 University of Utah CS interns; led UI/UX design in Figma alongside an intern designer, producing complete user flows and high-fidelity mockups

Nerd United Lehi, UT (Hybrid)
Blockchain Engineer April 2022 – April 2023

- Developed applications in Go and Solidity on EVM-based protocols; deployed and operated 8 Proof-of-Authority test networks and 4 production networks, and built a production Python pipeline orchestrating an image-generation API, IPFS pinning, and contract deployment as an automated workflow

Skullcandy Park City, UT
Firmware Architect October 2018 – March 2022

- Authored a C# desktop application for USB firmware updates and a Python CLI build toolchain with automated regression tests, replacing a manual process and cutting nightly validation cycles
- Architected embedded firmware across a portfolio of Bluetooth audio products, establishing platform-wide design patterns and coding standards that improved code quality and reduced per-product bring-up time
- Led a team of 3–5 engineers and worked daily with mechanical, electrical, manufacturing, and QA teams through the full product development cycle
- Served as primary liaison to Apple's MFi program office, translating technical constraints for executives, product managers, and non-technical partners

Code Corporation Draper, UT
Senior Firmware Engineer August 2013 – October 2018

- Led 3–4 engineers building advanced barcode reading systems from board bring-up through production release, driving Agile/Scrum adoption within the team
- Implemented a secure bootloader and Bluetooth Low Energy firmware for the Nordic nRF52832 SoC, and contributed Linux kernel patches for image sensor, GPIO/PWM, and USB Mass Storage support
- Integrated a JavaScript engine into device firmware for runtime-configurable behavior, letting customers change device logic in the field without a firmware release

TECHNICAL SKILLS

Frontend / Apps	Flutter & Dart (MVVM), React, Next.js, TypeScript, JavaScript, C# desktop applications, MUI, Tailwind CSS, responsive & cross-platform UI, Figma
Backend / APIs	Node/TypeScript, Python (FastAPI), ASP.NET Core, EF Core, REST & GraphQL API design, authentication & user accounts, serverless (AWS Lambda, AppSync, API Gateway), event-driven & real-time messaging
Databases	PostgreSQL (Supabase), relational schema & data modeling, T-SQL / stored procedures, DynamoDB
Hardware & Embedded	Bluetooth Low Energy (Nordic nRF52832), USB & serial communication, device & firmware protocols, secure bootloader, real-time embedded systems, Linux kernel drivers, board bring-up through production
Languages	C, C++, C#, TypeScript, JavaScript, Python, Dart, Go
Testing & Tooling	pytest, xUnit, Jest, React Testing Library, Playwright, automated regression testing, CI/CD, Git, code review, Agile/Scrum
AI-Assisted Dev	Claude Code in a spec/plan/review loop, Claude API (Anthropic SDK), structured outputs

PROJECTS

Shopify Inventory Sync & Storefront Alert ASP.NET Core 8, EF Core, T-SQL, xUnit
ASP.NET Core 8 / EF Core Web API that HMAC-verifies live inventory webhooks and persists updates through a T-SQL stored procedure; xUnit-tested and verified end-to-end against a real store. github.com/jsanders5/dotnet-shopify-demo

PySando: Interactive Python Learning Platform React, TypeScript, FastAPI, pytest
React frontend over a FastAPI backend that runs learner code in a sandboxed subprocess and grades it against hidden pytest suites, paired with an LLM tutor grounded in the live lesson and guardrailed to never reveal solutions. github.com/jsanders5/PySando

EDUCATION

University of Utah Salt Lake City, UT
M.S., Electrical & Computer Engineering • B.S., Computer Engineering 2013